

ters were:

$$\begin{aligned}\mu_1 &= [-3.88, 4]^\top, \quad \mu_2 = [3.04, -2.92]^\top, \\ \Sigma_1 &= \begin{bmatrix} 9.27 & 0.79 \\ 0.79 & 4.82 \end{bmatrix}, \quad \Sigma_2 = \begin{bmatrix} 2.87 & 0.03 \\ 0.03 & 3.78 \end{bmatrix}, \\ w_1 &= 0.49, \quad w_2 = 0.502.\end{aligned}$$

As Fig. 7 shows, LDA has not performed well enough as expected. The performance of QDA is more acceptable than LDA but still not good enough because QDA also assumes a uni-modal Gaussian for every class. The result of Gaussian naive Bayes is very different from the Bayes here because the Gaussian naive Bayes assumes uni-modal Gaussian with diagonal covariance for every class. Finally, the Bayes has the best result as it takes into account the multi-modality of the data and it is optimum (Mitchell, 1997).

13. Conclusion and Future Work

This paper was a tutorial paper for LDA and QDA as two fundamental classification methods. We explained the relations of these two methods with some other methods in machine learning, manifold (subspace) learning, metric learning, statistics, and statistical testing. Some simulations were also provided for better clarification.

This paper focused on LDA and QDA which are discriminators with one and two polynomial degrees of freedom, respectively. As the future work, we will work on a tutorial paper for non-linear discriminant analysis using kernels (Baudat & Anouar, 2000; Li et al., 2003; Lu et al., 2003), which is called *kernel discriminant analysis*, to have discriminators with more than two degrees of freedom.

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